

Problem 4

If a and b are complex numbers, show that the points $a - b$ and b are symmetric with respect to the point $\frac{a}{2}$.

Solution

We need to prove that $\frac{a}{2}$ is the midpoint of the segment connecting $a - b$ and b .

Definition of Symmetry

Two points z_1 and z_2 are **symmetric with respect to a point c** if c is the midpoint of the segment connecting z_1 and z_2 .

Equivalently, z_1 and z_2 are symmetric about c if:

$$c = \frac{z_1 + z_2}{2}$$

or equivalently:

$$z_1 + z_2 = 2c$$

Setup

Let:

- $z_1 = a - b$
- $z_2 = b$
- $c = \frac{a}{2}$

We need to show that c is the midpoint of z_1 and z_2 .

Proof Method 1: Direct Calculation of Midpoint

The midpoint of z_1 and z_2 is:

$$\frac{z_1 + z_2}{2} = \frac{(a - b) + b}{2} = \frac{a}{2}$$

This equals $c = \frac{a}{2}$.

Therefore, $\frac{a}{2}$ is indeed the midpoint of $(a - b)$ and b .

Proof Method 2: Verify Symmetry Condition

We need to verify that:

$$z_1 + z_2 = 2c$$

Computing the left side:

$$z_1 + z_2 = (a - b) + b = a$$

Computing the right side:

$$2c = 2 \cdot \frac{a}{2} = a$$

Since both sides equal a , we have $z_1 + z_2 = 2c$.

Proof Method 3: Equal Distances

Points z_1 and z_2 are symmetric about c if the distances from each to c are equal and they lie on opposite sides.

Distance from z_1 to c

$$|z_1 - c| = \left| (a - b) - \frac{a}{2} \right| = \left| \frac{a}{2} - b \right|$$

Distance from z_2 to c

$$|z_2 - c| = \left| b - \frac{a}{2} \right| = \left| \frac{a}{2} - b \right|$$

The distances are equal.

Collinearity and Opposition

We also need to verify that z_1 , c , and z_2 are collinear and c lies between them.

The vector from c to z_1 :

$$z_1 - c = (a - b) - \frac{a}{2} = \frac{a}{2} - b$$

The vector from c to z_2 :

$$z_2 - c = b - \frac{a}{2}$$

We can see that:

$$z_2 - c = -(z_1 - c)$$

This means the vectors point in opposite directions with equal magnitude, confirming that c is the midpoint.

Cartesian Coordinates Verification

Let $a = a_1 + ia_2$ and $b = b_1 + ib_2$.

Then:

- $z_1 = a - b = (a_1 - b_1) + i(a_2 - b_2)$ corresponds to point $(a_1 - b_1, a_2 - b_2)$
- $z_2 = b = b_1 + ib_2$ corresponds to point (b_1, b_2)
- $c = \frac{a}{2} = \frac{a_1}{2} + i\frac{a_2}{2}$ corresponds to point $(\frac{a_1}{2}, \frac{a_2}{2})$

The midpoint of $(a_1 - b_1, a_2 - b_2)$ and (b_1, b_2) is:

$$\left(\frac{(a_1 - b_1) + b_1}{2}, \frac{(a_2 - b_2) + b_2}{2} \right) = \left(\frac{a_1}{2}, \frac{a_2}{2} \right)$$

This equals the point corresponding to $c = \frac{a}{2}$.

Geometric Interpretation

In the complex plane:

- Point b is located at b
- Point $a - b$ is the result of translating the origin by vector a and then by vector $-b$
- Point $\frac{a}{2}$ is halfway along the vector from origin to a
- The three points form a line segment with $\frac{a}{2}$ exactly in the middle

Example

Let $a = 6 + 4i$ and $b = 2 + i$.

Then:

- $z_1 = a - b = (6 + 4i) - (2 + i) = 4 + 3i$ (point $(4, 3)$)
- $z_2 = b = 2 + i$ (point $(2, 1)$)
- $c = \frac{a}{2} = \frac{6+4i}{2} = 3 + 2i$ (point $(3, 2)$)

Midpoint of $(4, 3)$ and $(2, 1)$:

$$\left(\frac{4+2}{2}, \frac{3+1}{2} \right) = (3, 2) = c$$

Final Answer

Points $a - b$ and b are symmetric with respect to point $\frac{a}{2}$

Proof: The midpoint of $(a - b)$ and b is:

$$\frac{(a - b) + b}{2} = \frac{a}{2}$$

Therefore, $\frac{a}{2}$ is equidistant from both $(a - b)$ and b , and lies on the line segment connecting them, making the points symmetric about $\frac{a}{2}$.

Alternative form:

$$(a - b) + b = 2 \cdot \frac{a}{2}$$

which confirms the symmetry condition.